

Solar GARDEN LIGHT

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Most houses have backyard gardens or some open areas. Installing a garden light there provides visibility at night. But there can be problems in its installation since the wiring may hamper movement and disturb plants in the garden. Presented here is the circuit of a solar based light that is bright enough to illuminate a garden while not requiring any wiring to the grid supply and thus lowering the electricity bill.

Circuit and working

Circuit diagram of the solar garden light is shown in Fig. 1. It is built around a solar lamp controller IC CL0116 (IC1), a miniature solar cell, a bright white LED (LED1) and a few other components.

This circuit requires only a single Ni-Cd rechargeable battery to light up the white LED for more than five hours depending upon the ampere-hour (Ah) capacity of the battery.

When sunlight falls on the solar cell during daytime, the solar cell charges the rechargeable battery and turns LED1 'off'. When no light falls on the solar cell during nighttime, the IC draws power from the battery and provides constant current to light up LED1. Since the solar cell acts as a light sensor, there is no need of a photo-transistor or light-dependent resistor for automatic switching of the LED in the dark.

Lamp controller. IC CL0116 lamp controller is an application-specific integrated circuit (ASIC) in which solar charging and LED driving sections are integrated on the chip. It requires only an external inductor to construct a boost power supply with over 90 per cent efficiency. It offers the advantages of low power dissipation, low minimum operating voltage and

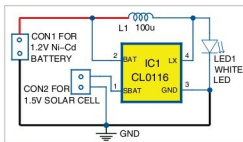


Fig. 1: Circuit diagram of solar garden light

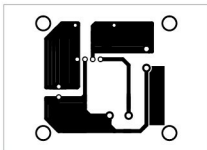


Fig. 2: Actual-size PCB layout of solar garden light

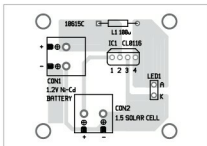


Fig. 3: Components layout for the PCB

PARTS LIST

Semiconductors:	
IC1	- CL0116 solar lamp controller
LED1	- 10mm bright white LED
Miscellaneous:	
CON1, CON2	- 2-pin terminal connector
L1	- 100µH inductor
	- 1.2V, 600mAh Ni-Cd battery
	- 1.5V, 200mA solar cell

automatic dusk-to-dawn switching.

IC CL0116 is available in TO-94 package and its overall circuit can be enclosed in a matchbox. There is no need of an external current-limiting resistor for the LED because the

LED driver circuit is integrated in the controller IC.

The IC operates off 0.8V to 3.0V. The voltage on SBAT pin should not go beyond 3.0V. For better protection, connect a 2.7V zener diode between SBAT pin and ground to maintain 2.7V if the solar cell voltage goes above 2.7V.

Solar cell. A 1.5V, 100mA solar cell is sufficient to charge the battery and light the LED. It can be mono-crystalline or polycrystalline. The nominal current of the solar cell should not be more than 800mA. For better performance, use a solar cell with nominal current below 400mA.

Construction and testing

An actual-size PCB layout for the solar garden light is shown in Fig. 2 and its components layout in Fig. 3.

Assemble the circuit, together with the LED, on the PCB. Use a wire length of not more than 5cm to connect LED1 to the IC. A ready-made inductor can be used for the circuit. For better performance and efficiency, use a low equivalent series resistance (ESR) value of inductor for the circuit. The circuit can fit in an old torchlight enclosure with appropriate modifications.

Fix connector CON1 and LED1 on the front, and CON2 at the top of the enclosure to connect the solar cell.

EFY notes. 1. Battery capacity of 600mAh to 1000mAh is large enough for this circuit.

2. In place of CL0116, you can use QX5252F, ANA608 or YX8018. **EFY**



A. Samiuddin is a B.Tech in electrical and electronics engineering with keen interest in LED lighting, power electronics, microcontrollers and Arduino programming